Wenjia Ge

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Research Interests

Human-in-the-loop and embodied AI systems; human-robot and human-AI interaction; large-scale audience behavior and content-platform experimentation; cross-cultural information behavior and technology use.

Education

New York University, Tandon School of Engineering *Sept 2024 – Dec 2026 (expected)*
M.S., Electrical Engineering, GPA: 3.93/4.0 | GRE: 327
Relevant coursework: Machine Learning, Interactive Medical Robotics, Systems Engineering, Real-Time Embedded Systems, Digital Signal Processing, Computing Systems Architecture, Applied Matrix Theory, Probability and Stochastic Processes

Shenzhen University, Shenzhen, China *Sept 2020 – July 2024*
B.S., Physics, GPA: 3.65/4.0 | Outstanding Undergraduate Thesis Award

Publications

Ge, W., Wen, P., Li, Z., Luo, Q., Zhao, C., Wang, J., & Yang, M. (2023). Application of the Method of Manufactured Solutions to Verify a Nuclear Reactor Physics Computational Code Using the Method of Characteristics. *Proceedings of the 30th International Conference on Nuclear Engineering (ICONE30)*, Paper 1702. doi:10.1299/jsmeicone.2023.30.1702

Research Experience

Research Assistant, AI4CE Lab (PI: Prof. Chen Feng) *Apr 2026 – Present*
New York University, New York, NY

- Collect and process real-world indoor and outdoor spatial data with MetaCam into 3D scene representations and point-cloud datasets for embodied-AI navigation research and world-model training; build preprocessing and quality-review pipelines.
- Build HPC workflows for MetaCam data ingestion, point-cloud preprocessing, 3D Gaussian Splatting reconstruction, and quality control for large-scale embodied navigation data.
- Student team lead for Wanderland, a multi-sensor capture and 3D-reconstruction pipeline for embodied-AI simulation that produced a CVPR 2026 Highlight publication; guide and onboard new students and track the team's data-collection progress.

Independent Research: Surgical Navigation Simulation *Feb – Aug 2026*
NYU (Advisor: Prof. Rui Li), New York, NY

- Designed and built a virtual simulation environment for a concentric-tube surgical robot, modeling robot mechanics and motion in constrained anatomical environments for minimally invasive surgery research.
- Built a real-time human-in-the-loop interaction pipeline connecting a haptic input device (3D Systems Touch) through ROS2 to RViz and 3D Slicer/SlicerROS2, enabling direct manual navigation and visualization within vascular anatomy derived from STL models, centerlines, and landmarks (VMTK).
- Owned hardware and driver integration, real-time visualization, and system validation; resolved instability issues from WSL2/USB passthrough.
- Worked on tip tracking, coordinate alignment, and constrained navigation for centerline-based progress measurement.

Independent Research: Audience Behavior & Content Experiments *Sept 2024 – Present*
Self-directed

- Design and run continuous content experiments (format, hook, posting cadence, visual direction) across social platforms reaching 150,000+ combined followers, using real-time engagement data to test hypotheses about user attention and retention.
- Grew a new account from 0 to 10,000 followers within one month through systematic testing of content variables.
- Translate complex technical and lifestyle topics into accessible content for international audiences.

Research Intern, Embodied AI and Robotics, AI2 Robotics *Oct 2025 – Jan 2026*
Shenzhen, China

- Independently validated system modules and evaluated their behavior across the perception–decision–action loop, designing test protocols and performance metrics across seven subsystems: perception, localization and mapping, task planning, motion control, manipulation, communications, and safety.
- Ran three technical demonstrations for prospective partners, contributing to new partnerships with organizations in Dubai and Singapore.
- Documented failure cases and supported cross-team troubleshooting across algorithm, hardware, and systems groups.

Undergraduate Thesis Research: Sb_2S_3 Thin-Film Solar Cells *2024*
Shenzhen University

- Fabricated FTO/CdS/Sb_2S_3/Spiro-OMeTAD/Au thin-film solar cells using chemical bath deposition and investigated thioacetamide surface treatment to improve absorber-layer and interface properties.
- Characterized crystallinity, surface morphology, wettability, and photovoltaic performance using XRD, SEM, contact-angle measurements, J-V testing, and EQE analysis, comparing treated devices with control samples.
- Improved power conversion efficiency from 4.43% to 5.43% through surface and interface modification; received the Outstanding Undergraduate Thesis Award.

Undergraduate Research: Nuclear Reactor Physics *2020 – 2024*
Shenzhen University

- Applied the Method of Manufactured Solutions to verify spatial-discretization accuracy of the Method of Characteristics in nuclear reactor physics simulation; conducted numerical modeling, coding, debugging, and verification.
- Resulted in a first-author publication at ICONE30.

Teaching Experience

Teaching Assistant, ECE 6113 / BE 6403 Digital Signal Processing I *Fall 2026*
NYU Tandon School of Engineering, New York, NY

- Grade student work and answer students' questions on Digital Signal Processing concepts.
- Coordinate weekly synchronous sessions; manage lecture recordings, Zoom logistics, and course material posting on Brightspace.

Graduate Student Mentor *Summer 2026*
NYU Tandon School of Engineering, New York, NY

- Advise incoming graduate students on course selection, academic planning, and campus and career resources.

Other Relevant Experience

Content & Communications, Kimber Health *July – Sept 2026*
New York, NY

- Conducted audience research on health-insurance information needs among international students at NYU, UCLA, and UChicago; developed localized content, FAQs, and in-person outreach materials to address the information gaps identified.
- Hosted and organized offline promotional events for the program.

Hardware Test Engineer Intern, Shenzhen Zhi'ao Technology Co., Ltd. *July – Aug 2023*
Shenzhen, China

- Conducted functional and performance testing on embedded motherboards and industrial PCs; contributed to a 15% reduction in hardware failure rates.

Honors and Awards

- Mentor Scholarship, NYU Tandon School of Engineering, Fall 2026
- Outstanding Undergraduate Thesis Award, Shenzhen University

Skills

Programming: Python, MATLAB, C, C++, Verilog, HTML/CSS/JavaScript; STM32 embedded development (ADC, DMA, timers, UART, SPI, I2C)
Robotics & Simulation: ROS2 (Jazzy), RViz, 3D Slicer, SlicerROS2, VMTK, MetaCam, 3D Systems Touch (haptic device)
Research Methods: audience and user behavior analysis, quantitative and qualitative data analysis, numerical verification methods
Languages: Mandarin Chinese (native), English (fluent)